LEAP: Lead Exposure Assistance Program

Quantitative Funding Framework: Technical Supplement

Municipal Infrastructure Finance Analysis

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1 Purpose of This Document

This document accompanies the five-phase quantitative presentation and assumes the reader has reviewed it. It does not restate phase-by-phase results or reproduce the recommendation tiers. Instead it provides four things the slides cannot adequately convey: the reasoning behind each distributional choice, a full account of the Phase 3 logistic regression, a structured analysis of the breakage rate uncertainty and what it would cost to reduce it, and a quantitative assessment of each model's directional biases.

2 Methodology Justification

2.1 Universe Estimation: Bayesian Conditional Probability

The cross-table approach treats the known-customer-material rows as a prior and applies the resulting hazard rate $\hat{p}(u)$ to the unknown-customer rows with the same utility material:

$$\hat{p}(u) = \frac{\displaystyle\sum_{j:\, U_j = u,\, C_j \in \mathcal{H}} n_j}{\displaystyle\sum_{j:\, U_j = u,\, C_j \neq \text{unknown}} n_j}$$

This is Bayesian in structure: we use observed co-occurrence of utility and customer materials to form beliefs about unobserved customer materials. The resulting rates and estimated hazardous properties are:

Utility Material	$\hat{p}(u)$	**Unknown Count**	**Est. Hazardous**
Lead / Galvanized	33.2%	14,097	4,701
Non-Lead Copper	9.6%	27,941	2,668
Unknown	16.8%	16,033	2,691
Other	0.1%	1,902	3
Total		59,973	**10,062**

$$N_{\text{total}} = 14{,}354 \text{ (confirmed)} + 10{,}062 \text{ (estimated)} = \mathbf{24{,}416}$$

The key identifying assumption is *missing at random conditional on utility material* — the reason a property's customer-side material is unrecorded is independent of what that material actually is, once we condition on utility-side material. If hazardous properties were systematically harder to inspect — because older neighborhoods had worse record-keeping — then $\hat{p}(u)$ would understate the true hazard rate and our universe estimate is a lower bound. This represents a structural downward bias in Phase 1 that is not testable with available data.

2.2 Replacement Cost: Log-Normal Fit and the KS Test

We fit a log-normal distribution to the 63 pilot contractor quotes using maximum likelihood estimation. The log-normal was chosen because replacement costs are the product of labor hours, material quantities, and contractor margin — a multiplicative generating process that the central limit theorem applied to logarithms predicts will be log-normally distributed. The fit produces:

Parameter	**Value**	**Implied Statistic**
$\hat{\mu}_C$ (log scale)	8.9065	Implied mean: \$7,603
$\hat{\sigma}_C$ (log scale)	0.2440	Implied median: \$7,380
KS statistic	0.1096	p-value: 0.41

The KS test computes the maximum absolute deviation between the empirical and fitted CDFs. A p-value of 0.41 means we cannot reject the log-normal hypothesis at any conventional significance level. To understand what rejection would have implied: a $p < 0.05$ would have indicated the distribution has features — bimodality, heavier tails, or a mass point — incompatible with a single log-normal family. In that case we would have considered a mixture of log-normals (capturing simple vs. complex replacements) or an empirical bootstrap. The high p-value makes these alternatives unnecessary, though with only 63 observations the test has limited power to detect moderate tail departures.

2.3 Participation Rate: Beta Distribution

The Beta(42, 40) distribution encodes the pilot outcome directly — 42 completions and 40 non-completions out of 82 applicants — as the conjugate posterior update to a uniform prior on $[0, 1]$:

$$\hat{\rho} \sim \text{Beta}(42, 40), \qquad \mathbb{E}[\hat{\rho}] = \frac{42}{82} = 51.22\%, \qquad 95\% \text{ CI} = [40.5\%,\ 61.9\%]$$

The width of this interval is the most important single fact about the model's uncertainty structure. The standard deviation of 5.5 percentage points, combined with a sensitivity of approximately \$2.2M in required capital per percentage point of participation, translates to a **\$12M standard deviation in capital requirements from participation uncertainty alone** — comparable to the entire interest rate sensitivity range in Phase 4.

The Beta distribution also has a meaningful limitation: it models participation as a fixed unknown rather than a time-varying quantity. In reality, participation is likely to increase as the 2037 mandate deadline approaches and enforcement pressure intensifies. A model allowing participation to drift upward over time would front-load uncertainty and change the shape of the cash flow distribution, though the 95th percentile of total required capital would likely be similar.

2.4 Sale Hazard: Why Weibull Over Exponential or Cox

The Weibull hazard function is:

$$h_{\text{sale}}(t) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1}$$

The shape parameter k directly determines whether hazard is increasing ($k > 1$), constant ($k = 1$, the exponential special case), or decreasing ($k < 1$). Our fitted value of $k = 1.1721$ from the Census B25026 Columbus tenure distribution confirms the hazard is *increasing* in tenure:

Parameter	**Value**	**Implication**
Shape k	1.1721	Increasing hazard — tenure raises sale probability
Scale λ	13.86 yr	Median tenure $\approx$ 10.1 years
Avg. implied hazard	7.90%/yr	vs. HMDA observed 4.78%

This is economically meaningful: homeowners who have lived in a property for 15 years are more likely to sell in any given year than those who have lived there for 2 years, reflecting life-cycle housing decisions (retirement, empty-nesting, estate settlement). An exponential model with constant hazard would miss this structure entirely.

The Cox proportional hazard model would have been a natural alternative, as it imposes no parametric form on the baseline hazard. However, Cox requires individual-level event time data, which we do not have — we have only the grouped tenure distribution from B25026. The Weibull is identifiable from grouped data; Cox is not. This makes Weibull the appropriate choice given our data constraints.

2.5 Interest Rate Model: Vasicek vs. Alternatives

The Vasicek model was chosen over geometric Brownian motion (GBM) and the Cox–Ingersoll–Ross (CIR) model:

$$dr_t = \kappa(\bar{r} - r_t)\,dt + \sigma_r\,dW_t$$

GBM is inappropriate for interest rates because it lacks mean reversion — rates historically do revert toward long-run equilibria, as the 2022–2024 hiking cycle and subsequent normalization demonstrate. Without mean reversion, GBM produces implausibly wide rate distributions over the 2026–2055 projection horizon. CIR improves on Vasicek by preventing negative rates through the $\sqrt{r_t}$ diffusion term, but requires additional estimation. We address the negative rate problem pragmatically by flooring at 0.5%, which is sufficient given that rate uncertainty contributes only \$12.4M to capital requirements — far less than the \$37.5M from breakage rate uncertainty. CIR's additional complexity is not warranted at this level of materiality.

The three scenario parameterizations are:

Scenario	r_0	$\bar{r}$	κ	σ_r
Low rates	5.25%	3.50%	0.40	0.60%
Base rates	5.25%	4.50%	0.30	0.80%
High rates	5.25%	5.50%	0.20	0.70%

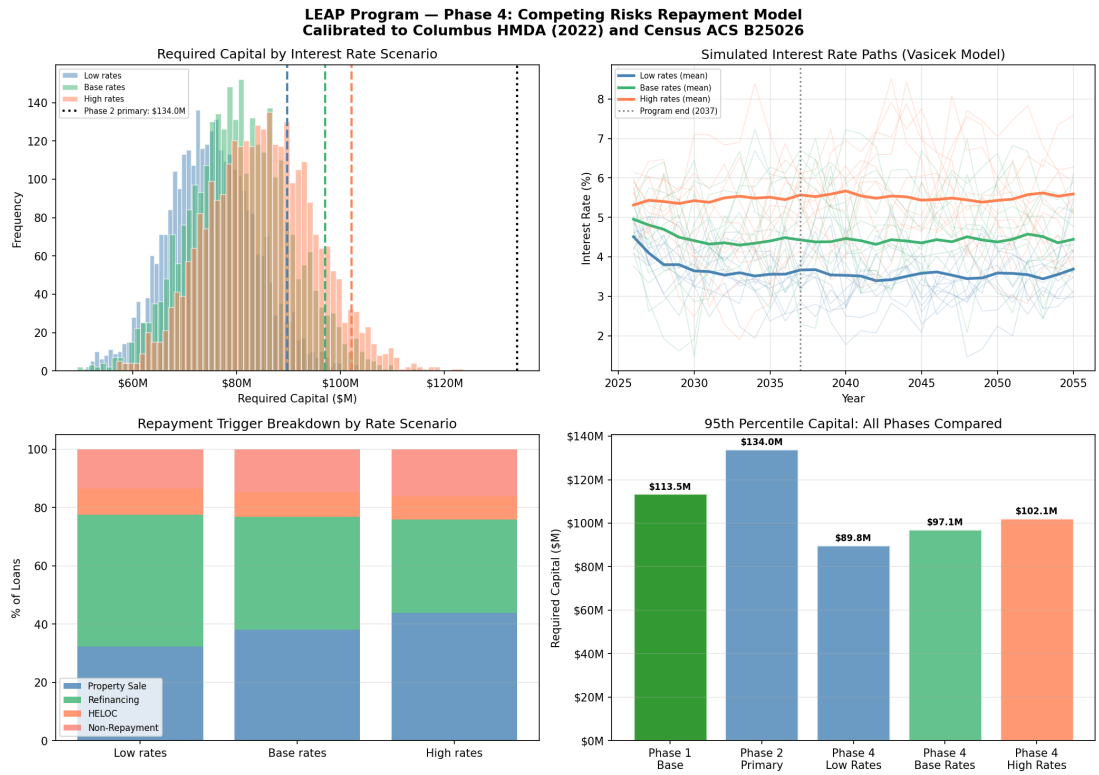

Figure 1: Phase 4 distributions. Notice that the distributions are generally close together despite ranging interest rates.

3 The Logistic Regression in Full

3.1 Model Specification and Fitted Coefficients

The Phase 3 participation model is:

$$P(\text{participate}_i = 1) = \sigma\left(\beta_0 + \sum_{j=1}^{5} \beta_j x_{ij}\right), \qquad \sigma(z) = \frac{1}{1+e^{-z}}$$

Because the outcome is binary and predictors include a mix of continuous and binary variables, logistic regression is preferred over a linear probability model (which can produce predictions outside $[0, 1]$) and over probit (which would produce nearly identical fitted probabilities but less interpretable coefficients in this setting). Features are standardized via z-score normalization before fitting. The fitted coefficients are:

Feature	Coefficient	Direction	Economic Interpretation
Owner-occupied	1.3615	↑	Strongest predictor; direct health benefit
Lead risk score	0.3180	↑	Higher severity → greater urgency
Assessed value	-0.2123	↓	Higher value → more likely to self-fund
Ownership length	0.3981	↑	Long tenure → attached to property
Year built	-0.2925	↓	Newer → less deteriorated pipes

All signs are consistent with prior domain knowledge. The model achieves a mean predicted participation rate of 51.61% against the pilot target of 51.22%, with a standard deviation of 0.271 across properties — indicating meaningful cross-sectional variation rather than a nearly-uniform prediction.

Predicted participation rates by property type reflect the model's structure: single-family homes (52.5%) outperform duplexes (50.4%) and multi-family properties (48.9%), primarily because the owner-occupancy rate is highest in single-family stock.

3.2 Multicollinearity Between Assessed Value and Ownership Length

Assessed value and ownership length are potentially collinear: properties held for many years are often in older neighborhoods with lower assessed values, and short-tenure properties in recently developed areas tend to have higher values. The practical consequence of collinearity is not bias in predicted probabilities — the model's predictions remain consistent — but inflated standard errors on individual coefficients, making it difficult to separately identify each factor's marginal effect.

For our purposes this is acceptable because we use the model for *prediction* (assigning property-level probabilities) rather than *causal inference* about which factor matters more for outreach design. If the goal were to design a targeted outreach campaign, we would address collinearity through lasso regularization or principal component regression to stabilize coefficient estimates.

3.3 External Validity

The model was trained on a synthetically generated dataset calibrated to Columbus demographics. Three features make it non-portable:

1. **Owner-occupancy rate.** The Columbus eligible population is 81% owner-occupied. In a city where hazardous lines are concentrated in rental stock — Baltimore, Cleveland — applying Columbus-calibrated coefficients to a fundamentally different property mix would systematically overestimate participation.

2. **Assessed value distribution.** The mean eligible property value in Columbus is $67,334. In higher-cost markets, the negative assessed-value coefficient would suppress predicted rates across the board, potentially underestimating uptake among properties that in Columbus would be mid-value but in those markets are low-value.

3. **Synthetic labels.** Because participation outcomes were generated from a latent score model rather than observed from actual enrollment, coefficient magnitudes encode our priors rather than empirical estimates. The sign pattern is correct; the magnitudes are not identified. A model trained on actual LEAP enrollment data as the program scales would be substantially more reliable.

4 Uncertainty Quantification: The $37.5M Breakage Gap

4.1 Structure of the Uncertainty

The breakage rate is the single largest source of model uncertainty in the analysis. The three scenarios and their 95th percentile capital requirements are:

Scenario	Rate	Derivation	95th Pct Capital
Primary	0.24%	59/24,416 pilot	$133,991,878
Adjusted	**1.38%**	$3.44\% \div 2.5$	**$164,705,237**
Adverse	3.44%	$(41/2862) \times (12/5)$	$218,243,730

The $37.5M gap between the primary recommendation ($165M) and the Phase 5 model-uncertainty-adjusted figure ($202.5M) arises from mixing these three distributions with different means ($110M, $137M, $186M) at weights 20/55/25. The mixture inflates the upper tail above what any single scenario produces:

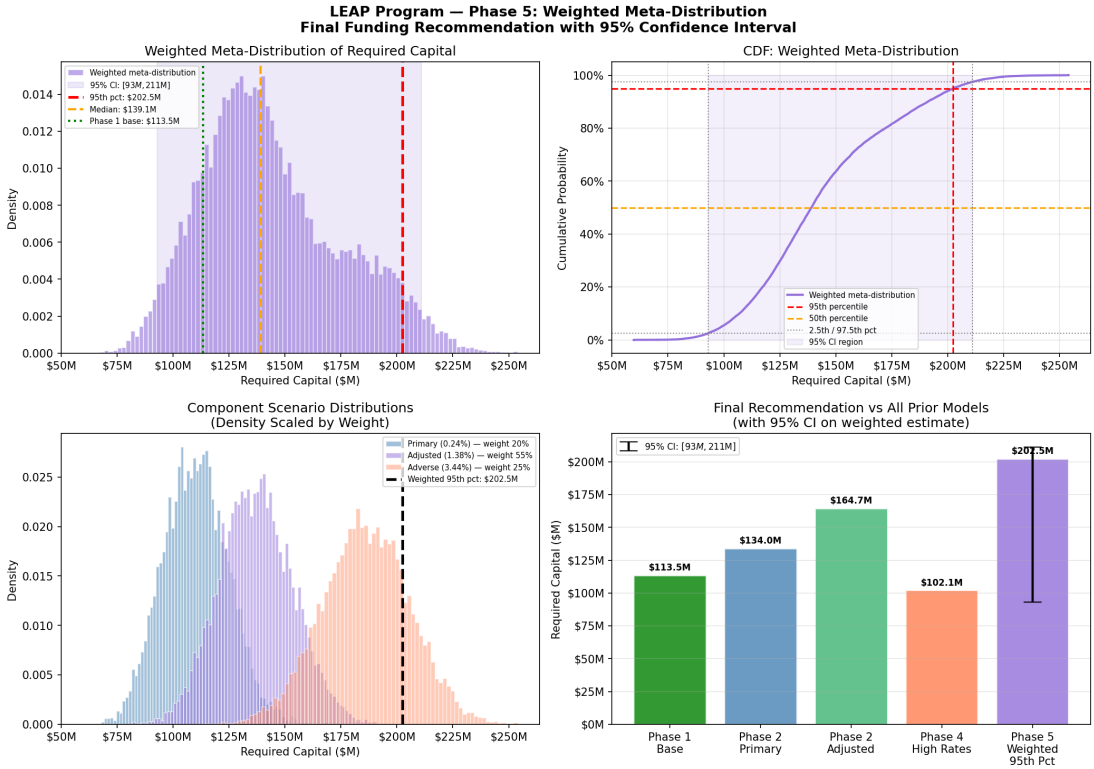

Figure 2: Phase 5 weighted meta-distribution. The 95th percentile of \$202.5M exceeds the adjusted scenario's own 95th percentile of \$164.7M because the adverse scenario's right tail inflates the mixture at the 95th percentile level. Bottom left shows the three component distributions at their true density scales.

This uncertainty is *reducible* — it is a measurement problem, not fundamental uncertainty about housing markets. The pilot rate is biased downward because the denominator is too large. The insurance rate is biased upward by adverse selection. Better measurement of the true population breakage rate could collapse both estimates toward a common value, dramatically narrowing the gap.

4.2 A Protocol for Resolution

A city finance officer seeking to reduce this uncertainty before committing the full capital reserve should consider:

Expanded monitoring cohort. In 2026–2027, actively track breakage events across a stratified random sample of 500–1,000 eligible properties, not just enrolled properties. With 1,000 monitored properties and a true rate of 1.38%, the expected number of events over two years is:

$$\mathbb{E}[\text{events}] = 1{,}000 \times 0.0138 \times 2 = 27.6$$

This yields a 95% CI of approximately $[18, 40]$ events, translating to a rate estimate of roughly $[0.9\%, 2.0\%]$ — a substantial tightening from the current $[0.24\%, 3.44\%]$ range.

Insurance data disaggregation. The 2,862 active WSL policies represent 11.72% of the eligible universe. If the insurer can provide claim rates stratified by property age, zip code, and pipe material, it becomes possible to estimate the selection bias factor empirically rather than assuming 2.5×. The current adjustment is:

$$\lambda_{\text{adjusted}} = \frac{\lambda_{\text{insurance}}}{\text{selection factor}} = \frac{3.44\%}{2.5} = 1.38\%$$

Even halving the uncertainty on the selection factor (from an assumed point estimate of 2.5 to a range of 2.0–3.0) would reduce the adjusted rate uncertainty from approximately 50% to 25% of the point estimate.

Cost-benefit. Each year of expanded monitoring costs approximately \$200,000–\$500,000. Against a potential \$37.5M reduction in required capital reserve, the benefit-cost ratio is roughly 75:1 to 100:1 — among the highest-return investments available to the program.

5 Limitations: Direction and Magnitude of Bias

5.1 Weibull Calibration from Grouped Census Data

The Weibull implies an average sale hazard of 7.90%/yr versus the HMDA-observed purchase rate of 4.78% — a 65% overestimate. This arises from fitting to Census B25026 bucket midpoints rather than the continuous tenure distribution. The midpoint assignment (e.g., all "5–14 year" owners assigned 9 years) understates within-bucket dispersion, particularly at the high-tenure end where the hazard is steepest.

Calibration	**Implied Avg Hazard**	**Bias Direction**
Weibull (grouped Census)	7.90%/yr	Overstates
HMDA observed purchase rate	4.78%/yr	Benchmark
Overshoot	**+3.12 pp**	**Understates capital**

The directional consequence: Phase 4 repayments arrive faster in simulation than they would in reality, causing Phase 4 to understate required capital. This reinforces rather than undermines the Phase 2 recommendation of \$165M, since Phase 2 does not use the Weibull.

5.2 HMDA Year Selection

The refi and HELOC rates are calibrated to 2024 HMDA data (mean refi rate 7.44%). In 2020–2021, national annual refi rates were 8–12%, approximately 4–5× higher than the 2024 rate of 2.33%. The Vasicek sensitivity function partially corrects for this, but the scaling is anchored to a single high-rate observation. The direction of bias is ambiguous: if long-

run equilibrium refi activity exceeds our modeled rates, Phase 4 understates repayments and overstates capital. This makes HMDA anchoring the most difficult limitation to sign a priori.

5.3 Independence of Competing Risks

Phase 4 treats sale, refi, and HELOC as independent:

$$P(T_{\text{sale}} = t \text{ and } T_{\text{refi}} = t) = P(T_{\text{sale}} = t) \times P(T_{\text{refi}} = t)$$

In reality these are correlated: a homeowner who refinances signals continued attachment and is less likely to sell in the following year. This causes the model to overcount repayment events, overstating repayment speed and understating required capital. The magnitude is likely small — competing events in the same year are relatively rare — but the bias directionally reinforces the Weibull overshoot.

5.4 Phase 2 Non-Repayment Omission

Phase 2 assumes 100% eventual repayment. Phase 3 identifies an 11.57% blended non-repayment rate. Integrating this into Phase 2 would increase required capital by approximately:

$$\Delta K \approx 0.1157 \times \$113.5\text{M} \approx \$13.1\text{M}$$

consistent with Phase 3's direct estimate of \$13.0M. The Phase 2 adjusted recommendation of \$165M is therefore understated by approximately \$13M, suggesting a fully integrated model would produce a primary recommendation closer to \$178M. This is the most material known bias in the analysis and the highest priority for future model integration.

5.5 Summary of Bias Directions

Limitation	**Direction**	**Approx. Magnitude**
Weibull overestimates sale hazard	Understates capital (Phase 4)	Material
HMDA high-rate anchor	Ambiguous	Unknown
Independent competing risks	Understates capital (Phase 4)	Small
Non-repayment omission (Phase 2)	Understates capital	≈\$13M
Missing-at-random universe estimate	Understates universe	Unknown

The preponderance of biases runs in the direction of understating required capital, which means the \$165M primary recommendation is more likely to be a lower bound on the truth than an overestimate. This further supports the Phase 5 model-uncertainty-adjusted figure of \$202.5M as the appropriate planning ceiling for a risk-averse city finance officer.

6 Synthesis

The five-phase framework produces estimates that are broadly consistent across approaches, which increases confidence in the \$165M–\$202.5M range. The directional biases identified above do not all point the same way: the Weibull overshoot, competing risks independence, and non-repayment omission all cause the model to understate required capital, while HMDA anchoring is ambiguous.

Three model refinements would most improve precision, in descending order of impact:

1. **Integrate Phase 3 non-repayment into Phase 2.** This single change raises the primary recommendation by ≈\$13M without requiring additional data and closes the largest known gap between the current model and a fully consistent framework.
2. **Individual-level deed transfer records for Weibull calibration.** Would resolve both the 65% hazard overshoot and the grouped-data bias in Phase 4, likely pushing Phase 4 estimates closer to the Phase 2 range.
3. **Two years of expanded breakage monitoring.** Could reduce the capital reserve requirement by up to \$37.5M — the highest-return data investment available to the program, with a benefit-cost ratio on the order of 75:1 to 100:1.